Exercise 13.1 SOLUTION

Q1. Find by definition, the derivatives w.r.t 'x' of the following function defined as.

(i) $2x^2+1$

Sol: Let $y = 2x^2+1$

Then $y+\delta y = 2(x+\delta x)^2+1$

$$\delta y = 2(x^2+2x\delta x+(\delta x)^2)+1-y$$

$$\delta y = 2x^2+4x\delta x+2(\delta x)^2+1-(2x^2+1)$$

$$\delta y = 2x^2+4x\delta x+2(\delta x)^2+1-2x^2-1$$

$$\delta y = 4x\delta x+2(\delta x)^2$$

$$\delta y = \delta x(4x+2\delta x)$$

Divide both side by δy and take limit $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}\frac{\delta x(4x+2\delta x)}{\delta x}\Rightarrow \frac{dy}{dx}=4x$$

نوٹ:

ان تمام سوالوں میں

1۔ دی گئی رقم کو y کے برابر لیں۔

2۔ x کی جگہ $x+\delta x$ اور y کی جگہ $y+\delta y$ لگائیں

3۔ y کو دائیں طرف لے جا کر منفی کریں اور پھر y کی جگہ اس کی قیمت درج کریں۔

4۔ تب تک حل کریں جب تک دائیں طرف سے δx مشترک نہ آجائے۔

5۔ دونوں اطراف δx کے ساتھ تقسیم کریں۔ اور limit لیں۔

(ii) $2-\sqrt{x}$

Sol: Let $y = 2-\sqrt{x}$

Then $y+\delta y = 2-\sqrt{x+\delta x}$

$$\delta y = 2-\sqrt{x+\delta x}-y = 2-\sqrt{x+\delta x}-2+\sqrt{x} = \sqrt{x}-\sqrt{x+\delta x}$$

$$\delta y = (\sqrt{x}-\sqrt{x+\delta x})\times\frac{\sqrt{x}+\sqrt{x+\delta x}}{\sqrt{x}+\sqrt{x+\delta x}} \quad \left(\begin{matrix}multiply\ and\ divide\ by\\ \sqrt{x}+\sqrt{x+\delta x}\end{matrix}\right)$$

$$\delta y = \frac{(\sqrt{x})^2-(\sqrt{x+\delta x})^2}{\sqrt{x}+\sqrt{x+\delta x}}=\frac{x-x-\delta x}{\sqrt{x}+\sqrt{x+\delta x}}$$

Divide by δx and take limit $\delta x\to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}\frac{-\delta x}{\delta x\left(\sqrt{x}+\sqrt{x+\delta x}\right)}=\lim_{\delta x\to 0}\frac{-1}{\left(\sqrt{x}+\sqrt{x+\delta x}\right)}$$

$$\frac{dy}{dx}=\frac{-1}{\sqrt{x}+\sqrt{x+0}}=\frac{-1}{\sqrt{x}+\sqrt{x}}=\frac{-1}{2\sqrt{x}}$$

(iii) $\dfrac{1}{\sqrt{x}}$

Sol: Let $y = \dfrac{1}{\sqrt{x}}$

Then $y+\delta y = \dfrac{1}{\sqrt{x+\delta x}}$

$$\delta y = \frac{1}{\sqrt{x+\delta x}}-y = \frac{1}{\sqrt{x+\delta x}}-\frac{1}{\sqrt{x}} = \frac{\sqrt{x}-\sqrt{x+\delta x}}{\sqrt{(x+\delta x)}\left(\sqrt{x}\right)}$$

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$$= \frac{\sqrt{x}-\sqrt{x+\delta x}}{\left(\sqrt{x+\delta x}\right)\sqrt{x}} \times \frac{\sqrt{x}+\sqrt{x+\delta x}}{\sqrt{x}+\sqrt{x+\delta x}} \qquad \begin{pmatrix} \text{multiply and divide by} \\ \sqrt{x}+\sqrt{x+\delta x} \end{pmatrix}$$

$$= \frac{\left(\sqrt{x}\right)^2-\left(\sqrt{x+\delta x}\right)^2}{\sqrt{x+\delta x}\sqrt{x}\left(\sqrt{x}+\sqrt{x+\delta x}\right)} = \frac{x-x-\delta x}{\sqrt{x+\delta x}\sqrt{x}\left(\sqrt{x}+\sqrt{x+\delta x}\right)}$$

Divide by δx and take limit $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{1}{\delta x}\frac{-\delta x}{\sqrt{x+\delta x}\sqrt{x}\left(\sqrt{x}+\sqrt{x+\delta x}\right)}$$

$$\frac{dx}{dy} = \lim_{\delta x\to 0}\frac{-1}{\sqrt{x+\delta x}\sqrt{x}\left(\sqrt{x}+\sqrt{x+\delta x}\right)} = \frac{-1}{\sqrt{x+0}\sqrt{x}\left(\sqrt{x}+\sqrt{x+0}\right)}$$

$$\frac{dy}{dx} = \frac{-1}{x(2\sqrt{x})} = -\frac{1}{2x^{3/2}}$$

(vi) $x(x-3)$

Sol: Let $y = x(x-3) = x^2-3x$

Then $y+\delta y = (x+\delta x)^2 - 3(x+\delta x)$

$$\delta y = x^2 + 2x\delta x + (\delta x)^2 - 3x - 3\delta x - y$$

$$= x^2 + 2x\delta x + (\delta x)^2 - 3x - 3\delta x - x^2 + 3x = \delta x(2x+\delta x-3)$$

Divide by δx and take limit $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{\delta x(2x+\delta x-3)}{\delta x}$$

$$\frac{dy}{dx} = 2x+0-3 = 2x-3$$

Q.2: Find $\frac{dy}{dx}$ from first principle and find gradient of the curve at the given point.

(i) $\sqrt{x+2}$ at $x=6$ (ii) $\frac{1}{\sqrt{x+a}}$ at $x=a$

(i) $\left(\sqrt{x+2}\right)$ at $x=6$

Sol: Let $y = (x+2)^{1/2}$

Then $y+\delta y = (x+\delta x+2)^{1/2}$

$\delta y = (x+2+\delta x)^{1/2} - y$

$\delta y = ((x+2)+\delta x)^{1/2} - (x+2)^{1/2} = ((x+2)+\delta x)^{1/2} - (x+2)^{1/2}$

$$= (x+2)^{1/2}\left[\left(1+\frac{\delta x}{x+2}\right)^{1/2}-1\right]$$

$$= (x+2)^{1/2}\left[1+\frac{1}{2}\left(\frac{\delta x}{x+2}\right)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+2}\right)^2+\ldots\ldots-1\right]$$

نوٹ:

ان سوالات میں جب x کی جگہ $x+\delta x$ لگائیں گے تو تین نمبر بن جائیں گے جبکہ Binomial کے لیے دو نمبر کی ضرورت ہوتی ہے تو δx کے علاوہ باقی دونوں نمبر کو ایک نمبر فرض کریں گے۔

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$$= (x+2)^{1/2} \cdot \left(\frac{\delta x}{x+2}\right)\left[\frac{1}{2} + \frac{\left(\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+2}\right) + \ldots\ldots\right]$$

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{(x-2)^{\frac{1}{2}-1} \cdot \cancel{\delta x}}{\cancel{\delta x}}\left[\frac{1}{2} + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+2}\right) + \ldots\ldots\right]$$

$$\Rightarrow \frac{dy}{dx} = (x+2)^{\frac{-1}{2}}\left(\frac{1}{2}+0\right) \qquad \Rightarrow \frac{dy}{dx} = \frac{1}{2}(x+2)^{\frac{-1}{2}} = \frac{1}{2\sqrt{x+2}}$$

$$\text{At } x = 6 \;, \quad m = \left.\frac{dy}{dx}\right|_{x=6} = \frac{1}{2}(6+2)^{\frac{-1}{2}} = \frac{1}{2}(8)^{\frac{-1}{2}} = \frac{1}{2\sqrt{8}} = \frac{1}{2\sqrt{4\times 2}} = \frac{1}{2\times 2\sqrt{2}} = \frac{1}{4\sqrt{2}}$$

(ii) $y = \dfrac{1}{\sqrt{x+a}} = (x+a)^{-1/2}$ at x = a

Sol:

$y + \delta y = (x + \delta x + a)^{-1/2}$

$\delta y = (x+a+\delta x)^{-1/2} - y = (x+a+\delta x)^{-1/2} - (x+a)^{-1/2}$

$$\delta y = (x+a)^{-1/2}\left[\left(1+\frac{\delta x}{x+a}\right)^{\frac{-1}{2}} - 1\right]$$

$$\frac{\delta y}{\delta x} = \frac{(x+a)^{\frac{-1}{2}}\left[1 + \left(-\frac{1}{2}\right)\left(\frac{\delta x}{x+a}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+a}\right)^2 + \ldots\ldots - 1\right]}{\delta x}$$

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{(x+a)^{\frac{-1}{2}} \cdot \frac{\cancel{\delta x}}{(x+a)}\left[-\frac{1}{2} + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}\left(\frac{\delta x}{x+a}\right) + \ldots\ldots\right]}{\cancel{\delta x}}$$

$$\Rightarrow \frac{dy}{dx} = (x+a)^{\frac{-3}{2}}\left(-\frac{1}{2}+0\right) \quad \Rightarrow \frac{dy}{dx} = -\frac{1}{2}(x+a)^{\frac{-3}{2}}$$

$$m = \left.\frac{dy}{dx}\right|_{x=a} = -\frac{1}{2}(a+a)^{\frac{-3}{2}} = -\frac{1}{2}(2a)^{\frac{-3}{2}} = -\frac{1}{2(2a)^{\frac{3}{2}}} = -\frac{1}{2}\cdot\frac{1}{2\sqrt{2}.a^{\frac{3}{2}}} = -\frac{1}{4\sqrt{2}a^{\frac{3}{2}}}$$

Q.3. i. Find the derivative of $x^{\frac{2}{3}}$ at $x = 8$ from the first principle.

Sol: Let $f(x) = x^{\frac{2}{3}}$.

$$f(x+\delta x) = (x+\delta x)^{\frac{2}{3}} = x^{\frac{2}{3}}\left(1+\frac{\delta x}{x}\right)^{\frac{2}{3}}$$

$$f(x+\delta x) - f(x) = x^{\frac{2}{3}}\left[1 + \left(\frac{2}{3}\right)\left(\frac{\delta x}{x}\right) + \frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \ldots\right] - x^{\frac{2}{3}}$$

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$$f(x+\delta x)-f(x)=x^{\frac{2}{3}}\left[1+\left(\frac{2}{3}\right)\left(\frac{\delta x}{x}\right)+\frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2+\ldots-1\right]$$

$$\frac{f(x+\delta x)-f(x)}{\delta x}=\frac{x^{\frac{2}{3}}\left(\frac{\delta x}{x}\right)\left[\left(\frac{2}{3}\right)+\frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\left(\frac{\delta x}{x}\right)+\ldots\right]}{\delta x}$$

$$\lim_{\delta x\to 0}\frac{f(x+\delta x)-f(x)}{\delta x}=\lim_{\delta x\to 0}x^{\frac{2}{3}-1}\left[\left(\frac{2}{3}\right)+\frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\left(\frac{\delta x}{x}\right)+\ldots\right]$$

$$f'(x)=x^{\frac{-1}{3}}\left(\left(\frac{2}{3}\right)+0+0+\ldots\right)\Rightarrow\boxed{f'(x)=\frac{2}{3}x^{\frac{-1}{3}}}$$

Put $x=8$

$$f'(8)=\frac{2}{3}8^{\frac{-1}{3}}=\frac{2}{3}(2)^{3\times\frac{-1}{3}}=\frac{2}{3}(2)^{-1}=\frac{2}{3}\times\frac{1}{2}=\frac{1}{3}$$

Q3.ii **Find the derivative of x^2+2x+3 by definition.**

Sol: Let $y=x^2+2x+3$.

(i) $y+\delta y=(x+\delta x)^2+2(x+\delta x)+3$ and

(ii) $\delta y=\left[(x+\delta x)^2+2(x+\delta x)+3\right]-\left[x^2+2x+3\right]$

$$\delta y=x^2+2x\delta x+5x^2+2x+2\delta+3-x^2-2x-3$$

(iii) $\dfrac{\delta y}{\delta x}=\dfrac{(\delta x)(2x+\delta x+2)}{(\delta x)}$

(iv) $\lim\limits_{\delta x\to 0}\dfrac{\delta y}{\delta x}=\lim\limits_{\delta x\to 0}(2x+\delta x+2)\Rightarrow\dfrac{dy}{dx}=2x+2$

Q.4. **Find from first principle, the derivatives of the following expressions w.r.t their respective independent variables**

(i) $(3x-2)^{-2}$

Sol: Let $y=(3x-2)^{-2}$

Then $y+\delta y=\left[3(x+\delta x)-2\right]^{-2}=(3x+3\delta x-2)^{-2}$

$$\delta y=(3x-2+3\delta x)^{-2}-(3x-2)^{-2}=(3x-2)^{-2}\left[\left(1+\frac{3\delta x}{3x-2}\right)^{-2}-1\right]$$

$$\delta y=(3x-2)^{-2}\left[1+\frac{(-2)3\delta x}{(3x-2)}+\frac{(-2)(-2-1)}{2!}\left(\frac{3\delta x}{3x-2}\right)^2+\ldots\ldots-1\right]$$

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$$\delta y=(3x-2)^{-2}\left[\frac{(-2)3\delta x}{(3x-2)}+\frac{(-2)(-3)}{2!}\left(\frac{3\delta x}{3x-2}\right)^2+.....\right]$$

$$\delta y=(3x-2)^{-2}.\left(\frac{3\delta x}{3x-2}\right)\left[(-2)+\frac{(-2)(-3)}{2!}\left(\frac{3\delta x}{3x-2}\right)+.....\right]$$

Divide by δx and take limit $\delta x\to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}\frac{(3x-2)^{-2-1}\,3\cancel{\delta x}\left[-2+\frac{(-2)(-3)}{2!}\left(\frac{3\delta x}{3x-2}\right)+.....\right]}{\cancel{\delta x}}$$

$$\Rightarrow\frac{dy}{dx}=(3x-2)^{-3}.3(-2+0)=-6(3x-2)^{-3}=\frac{-6}{(3x-2)^3}$$

(ii) $(2t+3)^5$

Sol: Let $y=(2t+3)^5$

Then $y+\delta y=\left[2(t+\delta t)+3\right]^5=(2t+2\,\delta t+3)^5=(2t+3+2\,\delta t)^5$

$$\delta y=(2t+3+2\,\delta t)^5-y=(2t+3+2\,\delta t)^5-(2t+3)^5$$

$$\delta y=(2t+3)^5\left[\left(1+\frac{2\delta t}{2t+3}\right)^5\right]-(2t+3)^5$$

$$\delta y=(2t+3)^5\left[\left(1+\frac{2\delta t}{2t+3}\right)^5-1\right]$$

$$\delta y=(2t+3)^5\left[\cancel{1}+5\left(\frac{2\delta t}{2t+3}\right)+\frac{5(5-1)}{2!}\left(\frac{2\delta t}{2t+3}\right)^2+.....-\cancel{1}\right]$$

$$\delta y=(2t+3)^5\left[5\left(\frac{2\delta t}{2t+3}\right)+\frac{5(4)}{2!}\left(\frac{2\delta t}{2t+3}\right)^2+.....\right]$$

$$\delta y=(2t+3)^5\left(\frac{2\delta t}{2t+3}\right)\left(5+\frac{5(4)}{2!}\left(\frac{2\delta t}{2t+3}\right)+.....\right)$$

Divide by δt and take lim $\delta t\to 0$

$$\lim_{\delta t\to 0}\frac{\delta y}{\delta t}=\lim_{\delta t\to 0}\frac{(2t+3)^{5-1}.2\cancel{\delta t}\left[5+\frac{5(4)}{2!}\left(\frac{2\delta t}{2t+3}\right)+.....\right]}{\cancel{\delta t}}$$

$$\frac{dy}{dt}=(2t+3)^4\,2(5+0)=5\times 2(2t+3)^4=10(2t+3)^4$$

(iii) $(aw+b)^7$

Sol: Let $y=(aw+b)^7$

$$y+\delta y=\left(a(w+\delta w)+b\right)^7$$

$$y+\delta y=(aw+a\delta w+b)^7$$

$$y+\delta y=(aw+b+a\delta w)^7$$

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$$\delta y = (aw+b+a\delta w)^7 - y$$

$$\delta y = (aw+b+a\delta w)^7 - (aw+b)^7$$

$$\delta y = (aw+b)^7\left(1+\frac{a\delta w}{aw+b}\right)^7 - (aw+b)^7$$

$$\delta y = (aw+b)^7\left[1+7\left(\frac{a\delta w}{aw+b}\right)+\frac{7(7-1)}{2!}\left(\frac{a\delta w}{aw+b}\right)^2+\ldots\ldots-1\right]$$

$$\delta y = (aw+b)^7.\left(\frac{a\delta w}{aw+b}\right)\left[7+\frac{7(6)}{2}\left(\frac{a\delta w}{aw+b}\right)+\ldots\right]$$

$$\lim_{\delta w\to 0}\frac{\delta y}{\delta w} = \lim_{\delta w\to 0}\frac{(aw+b)^6.a\delta w\left[7+7(3)\left(\frac{a\delta w}{aw+b}\right)+\ldots\right]}{a\delta w}$$

$$\frac{dy}{dw} = (aw+b)^6(a)(7+0)$$

$$\frac{dy}{dw} = 7a(aw+b)^6$$

Q.5. **Find the gradient and equation of the tangent line to $y=3x^2-4x+1$ at $x=2$.**

Sol: Given $y=3x^2-4x+1-------(i)$

$$\frac{dy}{dx} = \frac{d}{dx}(3x^2-4x+1)$$

$$\frac{dy}{dx} = 3(2x)-4(1)+0$$

$$\frac{dy}{dx} = 6x-4(1)+0 = 6x-4$$

$$m = \frac{dy}{dx}|_{x=2} = 6(2)-4$$

$$m = 12-4$$

$$m = 8$$

For point of tangency, put $x=2$ in (i)

$$y = 3(2)^2-4(2)+1$$

$$y = 3(4)-8+1$$

$$y = 12-8+1 = 5$$

The point of tangency is $(2,5)$.

Equation of tangent line is

$$y-y_1 = m(x-x_1)$$

Substituting $m=8$ and $(x_1,y_1)=(2,5)$:

$$y-5 = 8(x-2)$$

$$y-5 = 8x-16$$

$$y = 8x-16+5$$

$$y = 8x-11$$

The gradient of the tangent is 8 and the equation of tangent line is $y=8x-11$.

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Q.6. **For the function $f(x)=2x^3+x$, Calculate the equation of the tangent line at $x=-1$.**

Sol: Given $f(x)=2x^3+x$

$f'(x)=2.3x^2+1$

$f'(x)=6x^2+1$

$f'(-1)=6(-1)^2+1$

$f'(-1)=6(1)+1$

$f'(-1)=6+1$

$m=7$

$f(-1)=2(-1)^3+(-1)$

$f(-1)=2(-1)+-1$

$f(-1)=-2-1$

$f(-1)=-3$

The point of tangency is $(-1,-3)$

Equation of tangent line is $(y-y_1)=m(x-x_1)$

$y-(-3)=7(x-(-1))$

$y+3=7(x+1)$

$y+3=7x+7$

$y=7x+7-3$

$y=7x+4$

The equation of the tangent line is $y=7x+4$

Q.7. **Find the coordinates of the point of tangency and the equation of the tangent line for $f(x)=x^3-2x+1$ at $x=1$**

Sol: Given $f(x)=x^3-2x+1$

$f(1)=(1)^3-2(1)+1$

$f(1)=1-2+1$

$f(1)=0$

Take derivative w.r.t. x

$f'(x)=\frac{d}{dx}(x^3-2x+1)$

$f'(x)=3x^2-2$

Put $x=1$ to find $f'(1)$

$f'(1)=3(1)^2-2$

$f'(1)=3-2$

$f'(1)=1$

The slope of the tangent line is 1.

Equation of tangent line is $(y-y_1)=m(x-x_1)$

$y-0=1(x-1)$

$y=x-1$

The coordinates of the point of tangency are $(1,0)$ and the equation of the tangent line is $y=x-1$.

Q.8. **Find the gradient of the curve $f(x)=3x^2+2x$ at $x=1$**

Sol: Given $f(x)=3x^2+2x$

Take derivative w.r.t x

$f'(x)=3(2x)+2(1)=6x+2$

Put $x=1$

$f'(1)=6+2=8$

The gradient of the curve $f(x)=3x^2+2x$ at $x=1$ is 8.

Q.9. **Find the gradient and an equation of tangent line to the graph of $f(x)=\sqrt{x}$ at $x=9$.**

Sol: $f(x)=\sqrt{x}=x^{\frac{1}{2}}$ _____(i)

$f'(x)=\frac{1}{2}x^{-\frac{1}{2}}=\frac{1}{2\sqrt{x}}$

$f'(9)=\frac{1}{2\sqrt{9}}=\frac{1}{2.3}=\frac{1}{6}$

$f(9)=\sqrt{9}=3$

The point is $(9,3)$

Equation of tangent line is

$y-y_1=m(x-x_1)$

$y-3=\frac{1}{6}(x-9)$

$y=\frac{1}{6}x-\frac{9}{6}+3$

$x=\frac{1}{6}x-\frac{3}{2}+\frac{6}{2}$

$y=\frac{1}{6}x+\frac{3}{2}$

$6y=x+9$

The gradient of the tangent line is $\frac{1}{6}$ and the equation of the tangent line is $6y=x+9$

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Q.10. The position of a car after t hours is given by: $s(t) = 2t^3 - 3t^2 + t$ (in kilometers)

i. Find the average velocity over the interval $[1,4]$

ii. Find the instantaneous velocity at $t = 2$

Sol: Given $s(t) = 2t^3 - 3t^2 + t$

$$s(1) = 2(1)^3 - 3(1)^2 + 1 = 2 - 3 + 1 = 0$$

$$s(4) = 2(4)^3 - 3(4)^2 + 4 = 2(64) - 3(16) + 4 = 1$$

$$v_{avg} = \frac{s(4) - s(1)}{4 - 1}$$

$$v_{avg} = \frac{84 - 0}{3}$$

$$v_{avg} = 28$$

Instantaneous velocity

$$t = 2 : V'(2) = \lim_{\delta t \to 0} \frac{S(2+\delta t) - S(2)}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{S(2+\delta t)^3 - 3(2+\delta t)^2 + 2 + \delta t - \{2(2)^3 - 3(2)^2 + 2\}}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{16 + 2\delta t^3 + 24\delta t + 12\delta t^2 - 12 - 12\delta t - 3\delta t^2 + 2 + \delta t - 16 + 12 - 2}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{2\delta t^3 + 12st^2 + 13\delta t}{\delta t} = \lim_{\delta t \to 0} \frac{\delta t\left(2\delta t^2 + 12st + 13\right)}{\delta t}$$

$$= \lim_{\delta t \to 0} \left(2\delta t^2 + 12st + 13\right) = 13$$

The average velocity over the interval $[1,4]$ is 28 km/h and the instantaneous velocity at $t = 2$ is 13 km/h.

Q.11. A stone is thrown upwards and its height after t seconds is given by: $s(t) = -16t^2 + 32t + 10$ (in feet), Find the instantaneous velocity at $t = 1$.

Sol: $s(t) = 16t^2 + 32t + 10$

Instantaneous velocity $t = 1$ is

$$f'(1) = \lim_{\delta t \to 0} \frac{S(1+\delta t) - S(1)}{\delta t}$$

Now

$$S(1+\delta t) = -16(1+\delta t)^2 + 32(1+\delta t) + 10$$

$$= -16\left(1 + 2\delta t + \delta t^2\right) + 32 + 32\delta t + 10$$

$$= -16 - 32\delta t - 16\delta t^2 + 32 + 32\delta t + 10$$

$$= -16\delta t^2 + 26$$

$$S(1) = -16(1)^2 + 32(1) + 10 = 26$$

$$f'(1) = \lim_{\delta t \to 0} \frac{S(1+\delta t) - S(1)}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{-16\delta t^2 + 26 - 26}{\delta t}$$

$$= \lim_{\delta t \to 0} (-16\delta t) = 0$$

Hence, Instantaneous velocity is 0 at $t = 1$

Q.12. The outdoor temperature (in °C) over time is modeled by: $T(t) = -t^2 + 12t + 10$ where t is the time in hours. Find the instantaneous rate of change at $t = 2$.

Sol: Temperature function

$$T(t) = -t^2 + 12t + 10$$

The instantaneous rate of change at $t = 2$ is

$$T'(2) = \lim_{\delta t \to 0} \frac{T(2+\delta t) - T(2)}{\delta t}$$

$$T(2+\delta t) = -(2+\delta t)^2 + 12(2+\delta t) + 10$$

$$= -4 - \delta t^2 - 4\delta t + 24 + 12\delta t + 10$$

$$= -\delta t^2 + 8\delta t + 30$$

$$T(2) = -(2)^2 + 12(2) + 10$$

$$-4 + 24 + 10 = 30$$

$$T'(2) = \lim_{\delta t \to 0} \frac{T(2+\delta t) - T(2)}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{-\delta t^2 + 8\delta t + 30 - 30}{\delta t}$$

$$= \lim_{\delta t \to 0} \frac{\delta t\left(-\delta t + 8\right)}{\delta t} = 8$$

Thus, the instantaneous rate of change of the Temperature at $t = 2$ is 8° C/hour.

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